When is a logic algebraizable?

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Abstract

The phrase "algebraizable logic" ought to mean that a logic has good algebraic models. Intuitionistic multiplicative linear logic is a logic in its own right, with a cut-free sequent calculus and with models that are residuated commutative partially ordered monoids, and it is not algebraizable in the Blok–Pigozzi sense. The reason is the one Blok and Pigozzi gave for BCI: a defining equation must be built from terms of the language, and the language of the tensor, the linear implication and the multiplicative unit has no term expressing that a formula lies above that unit. Adding the modality removes the obstruction, because the modality supplies such a term, so the multiplicative exponential fragment comes out algebraizable. That verdict is an artefact of the signature. The models are the same monoids with an operator added, and the relation satisfying the definition is the external consequence relation rather than derivability itself. Passing from a symmetric monoidal closed category to its poset collapse leaves an order behind and no lattice, so the notion with an algebraic flavour here is order algebraizability over partially ordered monoids rather than algebraizability over varieties.

1 Introduction

Linear logic was given its algebraic models early and they have never been in doubt. A model of the intuitionistic multiplicative fragment is a symmetric monoidal closed category, and the proof-irrelevant version of that is a residuated commutative partially ordered monoid, which Martin Hyland and I called a lineale [14].[1] These are algebraic structures by any ordinary standard: a monoid, a residual, an order compatible with both. When a modality is added the structure acquires a unary operator, and the operator is as algebraic as the rest of it.

The literature on abstract algebraic logic nevertheless reports that linear logic is algebraizable only in forms that leave this fragment out. Galatos and Ono [11] prove that every substructural logic over **FL** is algebraizable in the Blok–Pigozzi sense [4], and Agliano [1] proves that MALL and LL are, with equivalent algebraic semantics the varieties of Girard algebras and of girales. The positive theorem there requires the language to contain the additive conjunction &, the unit and the implication. The multiplicative conjunction $\otimes$ does not serve in its place: the fragments shown there not to be algebraizable at all are ones having $\otimes$ and no &, the multiplicative fragment among them.

[1] Lineales are often taken for quantales, which they are not. A quantale is a complete lattice carrying an associative product that preserves arbitrary joins, and residuation there is a consequence of completeness rather than an axiom, since a map preserving all joins between complete lattices has a right adjoint. A lineale asks for a poset, asks for residuation outright, and asks for nothing further, so neither completeness nor the lattice operations are required. The difference is the one at issue throughout this note: a complete lattice has all meets and all joins, so calling a model of the multiplicative fragment a quantale hands back the additives that the fragment does not have, which is also why the phase semantics of linear logic always has them.

That verdict is a fact about the definition of algebraization rather than about the logic. The diagnosis I want to record here is due to Nick Galatos, who observed that where categorical semantics specialises to algebraic semantics it does so by taking the category to be a poset, and that a category with no products and no coproducts yields no meets and no joins on collapsing. Order algebraizability rather than algebraizability is accordingly the notion close to categorical semantics.

The two traditions are asking different questions. Algebraic logic asks which class of algebras stands to a given logic as Boolean algebras stand to classical propositional logic, and the reward for an answer is a supply of bridge theorems holding uniformly across a lattice of logics, graded by where a logic sits in the Leibniz hierarchy. The signature is fixed in advance, premisses are sets of formulas, and the generality comes from ranging over many logics at once. Categorical logic asks instead what structure the syntax of a logic carries and what its free model is, and there the generality comes from that structure being independent of any particular presentation, which is what Lawvere's reading of a theory as a category delivers. Neither question is a version of the other, and the word algebraizable names a classification in the first tradition where it would name an initiality statement in the second. For linear logic the gap matters more than usual, because the feature distinguishing the logic is the multiplicity of premisses, and that is what the first tradition abstracts away before the classification begins. Sections 2 to 4 fix the fragments, give their models and carry out the Blok–Pigozzi test, which sorts the two fragments in opposite directions; the later sections say what to put in its place.

2 Intuitionistic linear logic

The calculations below are about particular fragments, so the rules they range over are worth having on the page. Figure 1 gives the sequent calculus for intuitionistic linear logic, with sequents $\Gamma \Rightarrow A$ for Γ a finite multiset of formulas and A a single formula.

$$\frac{}{A \Rightarrow A}\ \mathrm{id} \qquad \frac{\Gamma \Rightarrow A \quad \Delta, A \Rightarrow B}{\Gamma, \Delta \Rightarrow B}\ \mathrm{cut}$$

$$\frac{\Gamma, A, B \Rightarrow C}{\Gamma, A \otimes B \Rightarrow C}\ \otimes L \qquad \frac{\Gamma \Rightarrow A \quad \Delta \Rightarrow B}{\Gamma, \Delta \Rightarrow A \otimes B}\ \otimes R \qquad \frac{\Gamma \Rightarrow C}{\Gamma, \mathbf{1} \Rightarrow C}\ \mathbf{1}L \qquad \frac{}{\Rightarrow \mathbf{1}}\ \mathbf{1}R$$

$$\frac{\Gamma \Rightarrow A \quad \Delta, B \Rightarrow C}{\Gamma, \Delta, A \multimap B \Rightarrow C}\ \multimap L \qquad \frac{\Gamma, A \Rightarrow B}{\Gamma \Rightarrow A \multimap B}\ \multimap R$$

$$\frac{\Gamma, A \Rightarrow C}{\Gamma, A \,\&\, B \Rightarrow C}\ \&\, L \qquad \frac{\Gamma, B \Rightarrow C}{\Gamma, A \,\&\, B \Rightarrow C}\ \&\, L \qquad \frac{\Gamma \Rightarrow A \quad \Gamma \Rightarrow B}{\Gamma \Rightarrow A \,\&\, B}\ \&\, R \qquad \frac{}{\Gamma \Rightarrow \top}\ \top R$$

$$\frac{\Gamma, A \Rightarrow C \quad \Gamma, B \Rightarrow C}{\Gamma, A \oplus B \Rightarrow C}\ \oplus L \qquad \frac{\Gamma \Rightarrow A}{\Gamma \Rightarrow A \oplus B}\ \oplus R \qquad \frac{\Gamma \Rightarrow B}{\Gamma \Rightarrow A \oplus B}\ \oplus R \qquad \frac{}{\Gamma, \mathbf{0} \Rightarrow C}\ \mathbf{0}L$$

$$\frac{\Gamma, A \Rightarrow C}{\Gamma, !\,A \Rightarrow C}\ \mathrm{der} \qquad \frac{!\,\Gamma \Rightarrow A}{!\,\Gamma \Rightarrow !\,A}\ \mathrm{pro} \qquad \frac{\Gamma \Rightarrow C}{\Gamma, !\,A \Rightarrow C}\ \mathrm{weak} \qquad \frac{\Gamma, !\,A, !\,A \Rightarrow C}{\Gamma, !\,A \Rightarrow C}\ \mathrm{contr}$$

Figure 1: Intuitionistic linear logic. Exchange is implicit in the use of multisets.

Two fragments are at issue. Intuitionistic multiplicative linear logic, IMLL, has the connectives $\otimes$, $\multimap$ and $\mathbf{1}$, so its rules are the first three displays of the figure. Adding the modality and its four rules gives IMELL. Neither fragment contains the additive conjunction, whose right rule is the one place in the figure where a context is shared between two premisses rather than split between them.

Weakening and contraction are available in IMELL *only on* !*-formulas.* The rules weak

and contr apply to a formula of the form $!A$ and to no other, which makes the modality the carrier of the structural rules and makes $!A$ a comonoid in the categorical models. Dereliction and promotion give the comonad and the two structural rules give the comonoid, so the lineale axioms of Section 3 are a transcription of these four rules and of nothing else.

A defining equation should be built from the connectives of the fragment. The equation $p \,\&\, \mathbf{1} \approx \mathbf{1}$, which is what the algebraizability theorems for substructural logics use, is not a statement in the language of IMELL, since the additive conjunction does not occur in the rules of that fragment. Using it to algebraize a fragment without the additive conjunction asks the semantics to be compositional on a connective the logic does not have, and what the result then reports is something about a larger language. Whatever is to serve as a defining equation for IMELL has to be written with $\otimes$, $\multimap$, $\mathbf{1}$ and $!$.

The base of this literature is Lambek's calculus with a lattice added. The full Lambek calculus, which is where the algebraizability and the interpolation results both live, has $\wedge$, $\vee$ and the four constants in its language, and its algebras are residuated lattices. Lambek's own calculus has none of these: its connectives are the product and the two residuals, and the categorial grammar tradition that follows him uses that language rather than the extended one. Intuitionistic multiplicative linear logic is the commutative Lambek calculus with a unit, so the fragments of Figure 1 without & are not an eccentric corner of the subject but the older system, and the word "full" records that a lattice has been adjoined to it. The additives then earn their keep differently in the two literatures: in the algebraizability theorems $p \,\&\, \mathbf{1} \approx \mathbf{1}$ encodes an order relation as an equation, while in the uniform interpolation proofs conjunction and disjunction assemble interpolants out of cases.

The fragment the theorems do cover is not a natural one. Agliano's positive result applies to any fragment containing $\{\multimap, \&, \mathbf{1}\}$, and the smallest of these, with $\top$ adjoined in the bounded case, is a fragment nobody would isolate on proof-theoretic grounds. Linear implication is the residual of the tensor, introduced by $\multimap R$ to internalise the comma on the left of a sequent, and in a fragment with no tensor there is no connective corresponding to that comma, so the implication is left without the operation it is adjoint to. Agliano's own analysis bears this out. The equivalent algebraic semantics he computes for $\{\multimap, \&, \mathbf{1}\}$ is the quasivariety of Girard semilattices, which is not a variety, in which the implication is a BCI implication and not a residuation; and the positive fragment of MALL is not a conservative extension of it, the $\{\multimap, \&, \mathbf{1}\}$-subreducts of commutative residuated lattices forming a proper subvariety. The fragment where algebraizability goes through is therefore one whose implication has lost the adjunction that makes it linear implication.

3 Lineales

The shape of the models decides which notion of algebraizability can fit them, so they come before the test. The categorical models of IMELL are symmetric monoidal closed categories $\mathcal{V}$ equipped with a comonad for the modality. Collapsing $\mathcal{V}$ to a poset, by identifying objects with the same hom-behaviour and reading $a \leq b$ as the existence of a map from a to b, leaves a lineale with a unary operator on it.

Definition 1. A *lineale* is a structure $\langle L, \leq, \otimes, 1, \multimap \rangle$ where $\langle L, \leq \rangle$ is a poset, $\langle L, \otimes, 1 \rangle$ is a commutative monoid with $\otimes$ monotone in each argument, and residuation holds: for all $a, b, c \in L$,

$$a \otimes b \leq c \quad \Longleftrightarrow \quad a \leq b \multimap c.$$

A *modality* on a lineale is a monotone map $!$ satisfying dereliction $!\,a \leq a$, promotion $!\,a \leq !\,!\,a$, the unit condition $1 \leq !\,1$, and the two comonoid conditions $!\,a \leq 1$ and $!\,a \leq !\,a \otimes !\,a$.

A lineale has no meets. Definition 1 asks for a partial order and not for a lattice, and this is not an oversight to be repaired: a symmetric monoidal closed category need have no products, and a lineale that happens to have binary meets is carrying the additive conjunction as well as the multiplicative one. The same applies to the involution. Girard algebras are commutative residuated lattices with an involutive 0, so that notion has two pieces of extra structure built in before any multiplicative content is reached, and the absence of an involution in a lineale is a feature of the setting rather than a deficiency of it.

The comonoid conditions replace the axiom that needs meets. The usual axiomatisation of a storage operator, in the form Ono gives it, asks for $!\,1 = 1$, $!\,a \leq a\&1$, $!\,a\,!\,b = !(a\&b)$ and $!\,!\,a = !\,a$. Two of these come across unchanged. The second splits into $!\,a \leq a$ and $!\,a \leq 1$, neither of which mentions a meet, and the fourth is automatic in a poset, since promotion gives $!\,a \leq !\,!\,a$ while dereliction applied at $!\,a$ gives $!\,!\,a \leq !\,a$. Only the third genuinely needs the additive conjunction, and in a lineale its work is done by the two comonoid conditions, which make $!\,a$ a comonoid and so make the rules weak and contr available on !-formulas.

One consequence of these axioms is used repeatedly below, so it is recorded separately.

Lemma 2. *Let L be a lineale with a modality and $a \in L$. Then $!\,a = 1$ if and only if $1 \leq a$.*

Proof. Suppose $1 \leq a$. Monotonicity of ! gives $!\,1 \leq !\,a$, and the unit condition $1 \leq !\,1$ gives $1 \leq !\,a$; with the comonoid condition $!\,a \leq 1$ this yields $!\,a = 1$. Conversely, suppose $!\,a = 1$. Dereliction gives $!\,a \leq a$, so $1 \leq a$. □

4 Algebraizability

The notion being tested is the one Blok and Pigozzi defined, and the test turns on the shape of the definition rather than on any difficulty in finding a class of algebras, so it needs stating precisely.

Definition 3. A *deductive system* S over a propositional language L is a relation $\vdash_S \subseteq \mathcal{P}(Fm_L) \times Fm_L$ that is reflexive, monotone, transitive and invariant under substitution. Such a system is *algebraizable* with equivalent algebraic semantics a quasivariety K when there are a finite set of equations $\delta_i(p) \approx \varepsilon_i(p)$ in one variable, the defining equations, and a finite set of formulas $\Delta_j(p, q)$ in two variables, the equivalence formulas, such that for all $\Gamma \cup \{\varphi\} \subseteq Fm_L$

$$\Gamma \vdash_S \varphi \quad \Longleftrightarrow \quad \{\delta(\gamma) \approx \varepsilon(\gamma) : \gamma \in \Gamma\} \models_K \delta(\varphi) \approx \varepsilon(\varphi)$$

and $p \approx q \models|_K \delta(\Delta(p, q)) \approx \varepsilon(\Delta(p, q))$.

The two clauses together amount to a lattice isomorphism between the theories of S and the K-congruences, commuting with substitution.

A defining equation is a term saying that a formula is a theorem. The first clause of Definition 3, read with Γ empty, requires $\vdash_S \varphi$ to hold exactly when $\delta(\varphi) \approx \varepsilon(\varphi)$ holds throughout K, so the language must contain terms expressing theoremhood as an equation between them. Where the intended models are ordered and the theorems are the formulas denoting elements above the unit, as in every logic discussed here, this comes to asking for a term that expresses $1 \leq p$.

Proposition 4 (Blok–Pigozzi)**.** *Intuitionistic multiplicative linear logic is not algebraizable in the sense of Definition 3.*

Blok and Pigozzi established this for BCI, the implicational fragment, and then for the $\{\multimap, \otimes\}$-fragment [4, 5]; the argument extends to the whole multiplicative fragment. It

runs through matrices, a matrix being an algebra together with a distinguished subset of its elements, the designated ones, and being a model of a deductive system when the rules carry designated premisses to designated conclusions; it is reduced when the identity is the only congruence leaving the designated set invariant, and the algebras eligible to serve as equivalent algebraic semantics are the algebra parts of the reduced models. For the multiplicative fragment those models are not algebras with a subset. They are commutative partially ordered residuated semigroups, each carrying the filter generated by its elements $a \multimap a$, and no term of the language defines the partial order, which is therefore extra structure that no quasivariety can absorb.

Proposition 5. *The external consequence relation of intuitionistic multiplicative exponential linear logic is algebraizable in the sense of Definition 3, with defining equation* $!\,p \approx \mathbf{1}$ *and equivalence formulas* $\{p \multimap q,\ q \multimap p\}$.

Proof. By the criterion of Blok and Pigozzi [4] it is enough to check that $\Delta = \{p \multimap q, q \multimap p\}$ is a set of equivalence formulas for $\vdash^E$ and that $p \dashv\vdash^E \Delta(!\,p, \mathbf{1})$. Every derivation below is in the calculus of Figure 1, with each premiss adjoined as an axiom $\Rightarrow \gamma$.

For the equivalence formulas, reflexivity is id followed by $\multimap R$, symmetry holds because Δ is closed under exchanging its two members, and transitivity is cut. Congruence for $\otimes$ and for $\multimap$ is the functoriality of those connectives and needs nothing beyond IMLL. For the modality, assume $\Rightarrow p \multimap q$. From $p \Rightarrow p$ and $q \Rightarrow q$ the rule $\multimap L$ gives $p, p \multimap q \Rightarrow q$, so cut against the assumption gives $p \Rightarrow q$; dereliction gives $!\,p \Rightarrow q$, and promotion applies because the context is $!\,p$, so $!\,p \Rightarrow !\,q$ and $\multimap R$ finishes. The other member of Δ is symmetric.

For the defining equation, one half needs no assumption: the rule $\mathbf{1}R$ gives $\Rightarrow \mathbf{1}$, weakening gives $!\,p \Rightarrow \mathbf{1}$, and $\multimap R$ gives $\Rightarrow !\,p \multimap \mathbf{1}$. For the other half, assume $\Rightarrow p$. Promotion applies because the context is empty, giving $\Rightarrow !\,p$; then $\mathbf{1}L$ gives $\mathbf{1} \Rightarrow !\,p$ and $\multimap R$ gives $\Rightarrow \mathbf{1} \multimap !\,p$. Conversely, assume both members of $\Delta(!\,p, \mathbf{1})$. Dereliction on $p \Rightarrow p$ gives $!\,p \Rightarrow p$, which with $\Rightarrow \mathbf{1}$ and $\multimap L$ gives $\mathbf{1} \multimap !\,p \Rightarrow p$, and cut against the assumption $\Rightarrow \mathbf{1} \multimap !\,p$ gives $\Rightarrow p$. □

By Lemma 2 the equation $!\,p \approx \mathbf{1}$ expresses $1 \leq p$ in the language of the fragment, which is what Proposition 4 shows to be unavailable without the modality. The step carrying that half of the proof is promotion on an empty context, and the same step has a consequence recorded in the next paragraph but one.

The modality rescues the algebraization because it carries the structural rules. Among the rules of Figure 1 the modality is distinguished by weak and contr, which apply to a formula $!\,A$ and to no other, so ! is the device by which IMELL readmits weakening and contraction on marked formulas. That same marking is what supplies the missing term, since $!\,a = 1$ holds exactly for the elements above the unit. The connective whose proof-theoretic role is to restore the structural rules is thus the one that makes the algebraization go through, and the line separating Proposition 4 from Proposition 5 tracks the presence of that device rather than any difference in the models, which are the lineales of Definition 1 with and without an operator.

Which consequence relation is algebraized is a further question. Derivability in IMELL relates a finite multiset of formulas to a formula, and multiplicity matters there, since $p, p \Rightarrow p \otimes p$ is derivable while $p \Rightarrow p \otimes p$ is not; a relation whose left-hand side is a set cannot record the difference. Avron [3] separates this internal relation from the external one, $\Gamma \vdash^E \varphi$ holding when $\Rightarrow \varphi$ is derivable in the calculus with $\Rightarrow \gamma$ adjoined as an axiom for each $\gamma \in \Gamma$, and Proposition 5 is stated for the external relation because the internal one is not a deductive system in the sense of Definition 3 at all. The external relation cannot tell p from $!\,p$ either: promotion on an empty context gives $p \vdash^E !\,p$ and dereliction gives the converse, which is also what Agliano's deduction theorem for linear logic, $\Gamma, p \vdash q$ if and only

if $\Gamma \vdash {!}p \multimap q$, says when Γ is empty and q is ${!}p$. The defining equation of Proposition 5 therefore works because the relation being algebraized has already identified a hypothesis with its !. The modality is not thereby trivial in the models, where a lineale or a girale carries a perfectly nontrivial one; the collapse is in the consequence relation.

5 Order algebraizability

With the shape of the models fixed, the question becomes which abstract framework takes structures of that shape as its semantics. Blok–Pigozzi algebraizability asks for mutually inverse translations between the consequence relation of a logic and the equational consequence of a class of algebras, and the defining equation used for substructural logics over **FL** is $p \mathbin{\&} 1 \approx 1$. That equation writes $1 \leq p$ using an operation, and the additive conjunction appears in the positive theorems for exactly this reason, which is why the resulting classification tracks which connectives happen to sit in a fragment rather than what the fragment proves.[2]

Inequations are the natural target, not equations. Raftery's order algebraizable logics [19] replace equational consequence by inequational consequence over partially ordered algebras. The ordered model class of a logic is not unique, so the definition is relativised to a polarity on the signature, recording for each argument of each connective whether it is monotone or antitone, and each deductive filter then carries a Leibniz order in place of a Leibniz congruence. A lineale comes with an evident polarity, with $\otimes$ monotone in both arguments and $\multimap$ antitone in the first and monotone in the second, so the relativisation is fixed by the connectives rather than left as a free parameter. On the algebraic side the matching universal algebra is the theory of po-varieties, classes of partially ordered structures defined by inequations, which is the setting Jipsen and his coauthors work in for residuated posets [7].

6 The internal relation

The order of a lineale is not merely a convenience for stating the axioms: it is the consequence relation of the logic, and the derivation showing this is what ties the previous two sections together.

The order of a lineale is the internal consequence relation. Taking $\Gamma = [\varphi_1, \ldots, \varphi_n]$ a finite multiset, the internal relation holds between Γ and ψ when $\Gamma \Rightarrow \psi$ is derivable, which in a lineale reads $1 \leq (\varphi_1 \otimes \cdots \otimes \varphi_n) \multimap \psi$. Residuation with $a = 1$ gives $1 \otimes b \leq c$ if and only if $1 \leq b \multimap c$, and $1 \otimes b = b$, so that $b \leq c$ if and only if $1 \leq b \multimap c$. Putting $b = \varphi_1 \otimes \cdots \otimes \varphi_n$ and $c = \psi$ turns the internal relation into

$$\varphi_1 \otimes \cdots \otimes \varphi_n \;\leq\; \psi,$$

with the empty multiset sent to 1. The tensor absorbs the multiset structure of the premisses and only the order is left.

This reconciles the two complaints of Section 4. Work on multiset consequence relations, from Blok and Jónsson [6] through Galatos and Tsinakis [12] to Cintula, Gil-Férez, Moraschini and Paoli [8], builds the apparatus for premisses that form a multiset rather than a

[2] Making the indexing categorical does not address this. Voutsadakis's categorical abstract algebraic logic [20, 21] redoes the Leibniz hierarchy over π-institutions, the deductive counterpart of institutions due to Fiadeiro and Sernadas [9], in which the model and satisfaction components are replaced by a closure operator on subsets of the sentences of each signature. The consequence component is therefore Tarskian, and weakening and contraction are built in as before.

set, and the last of these reads the multiset relation exactly as a product-and-order statement in the commutative integral case. At poset level the objection about the relation and the objection about the connectives therefore turn out to be the same objection, and order algebraizability carries both. The non-integral case is not yet covered, since a lineale need not have 1 as its top element, and neither is the modality, which does not appear in that line of work at all.

7 Two conditions

What should replace Definition 3 is a requirement on the Lindenbaum construction together with a test on the notion that requirement produces. Both are stated here and then discussed in turn.

Proposal 6. Call a logic *algebraized* by a class of models when the Lindenbaum construction on a set of generators exhibits the syntactic category of the logic as the free model on those generators, so that interpretation in any model is the unique map out of it. Call the resulting notion *adequate* when the bridge theorems follow from it, interpolation against amalgamation and Beth definability against surjectivity of epimorphisms among them.

The first condition has to be freeness and not completeness. Every logic is complete with respect to its own Lindenbaum matrices, which is precisely why Blok and Pigozzi asked for the translations of Definition 3 to be mutually inverse rather than merely for models to exist. Soundness and completeness by Lindenbaum–Tarski is the right requirement, but only when the construction is asked to do more than produce some model. For IMELL the demand is that the construction yields the free symmetric monoidal closed category with a modality, and the statement is one of initiality. It is available before the poset collapse as well as after it, and at poset level it specialises to the lineale generated by the formulas under the internal relation, so the proof-relevant and proof-irrelevant versions are instances of one requirement rather than two separate theories.

The second condition is a test and not a clause. Amalgamation fails for many varieties of residuated lattices, so a definition that required it would exclude structures that are perfectly algebraic. What the test checks is that a candidate notion has consequences, and the next section is about where those consequences are already available.

8 Interpolation

The bridge theorems of Proposal 6 have been proved categorically for the intuitionistic case, and the forms they take there are worth setting out, because several of them cannot be stated in the equational framework at all.

Amalgamation and interpolation are a statement about pushouts. Pitts [15] proved the amalgamation property for Heyting algebras by categorical means, and derived interpolation for intuitionistic propositional logic from it; the companion paper of the same year carries the argument through open maps of the dual spaces. The pretopos version [16] proves a Craig interpolation theorem with Beth definability as a corollary, and conceptual completeness alongside it, which is where the Beth half of the bridge comes from in the categorical setting. There is also a proof-relevant form with no equational counterpart, that proofs factor through their interpolants, so that pushouts of bicartesian closed categories interpolate rather than merely their poset collapses.

Uniform interpolation is an adjointness. Uniform interpolation asks the interpolant to depend on one side only, so that a formula α has a strongest consequence α^* in a restricted

set of variables and a formula β a weakest antecedent β^*, and those two are the adjoints: writing ι for the inclusion of the free algebra on a set of variables into the free algebra on one variable more, α^* is the least element above α in the image of ι and β^* the greatest below β, which are the values of the left and the right adjoint. Pitts [17] proved that second-order propositional quantification can be interpreted in intuitionistic propositional logic, which says that the inclusion of the free Heyting algebra on n generators into the free Heyting algebra on $n+1$ has both adjoints. His proof analyses a terminating sequent calculus; Ghilardi and Zawadowski [13] prove the same result through a sheaf-theoretic representation of finitely presented Heyting algebras, and connect it to the model theory of those algebras, uniform interpolation corresponding to the existence of a model completion for the theory. The free algebras here are not complete, so the adjoints cannot be defined by infinite meets and joins, which is what makes the statement delicate and what the sheaf representation supplies.

For linear logic the interpolation results stop short of the adjointness. Craig interpolation for classical linear logic is old, proved syntactically from cut-free calculi by Roorda [18], and the intuitionistic case would need that work redone. On the algebraic side Fussner and Santschi [10] do treat the modality, working with girales and with a guarded interpolation property for formulas $!\varphi \multimap !\psi$, and they exhibit continuum-many extensions of linear logic with deductive interpolation, none of which has Craig interpolation. Neither line gives the adjoint statement, which for the linear case presupposes a categorical model of first-order intuitionistic linear logic.

The first-order case tells against the intuitionistic setting rather than for it. Blok–Pigozzi algebraizability is a propositional notion whose first-order surrogate runs through cylindric and polyadic algebras, whereas Craig interpolation together with conceptual completeness lifts to hyperdoctrines and pretoposes without being rebuilt. Uniform interpolation behaves differently. Henkin showed that classical and intuitionistic predicate logic both lack it, while Alizadeh, Derakhshan and Ono [2] prove it for the predicate extensions of the basic substructural logics, by Pitts's method, which needs every rule to reduce weight strictly and so cannot tolerate contraction. Their fragments carry no modality, and the rule contr of Figure 1 is a contraction, so IMELL lies outside the method and not merely outside the published results. Their interpolants are definable propositional quantifiers, which is the syntactic form of the adjointness; that adjointness has not been stated for substructural logics, and neither has the categorical version.

9 Conclusions

Propositions 4 and 5 put the two fragments of Figure 1 without an additive conjunction on opposite sides of the Blok–Pigozzi definition. Intuitionistic multiplicative linear logic is excluded, for the reason Blok and Pigozzi gave for BCI, because no term of its language expresses that a formula lies above the multiplicative unit. Intuitionistic multiplicative exponential linear logic is admitted, because the modality supplies such a term by Lemma 2, and admitted on its external consequence relation rather than on derivability, a relation that identifies a hypothesis with its !. The connective responsible for the second verdict is the one carrying weakening and contraction, so what the definition registers is the presence of a device for readmitting the structural rules, and the models on the two sides are the lineales of Definition 1 with and without an operator. The fragment where the published theorems do apply, with linear implication and the additive conjunction but no tensor, has an implication that is not a residuation, so algebraizability there is not a result about linear implication either.

Against this, Proposal 6 asks for freeness of the Lindenbaum construction and tests the resulting notion by the bridge theorems, which are available in categorical form for the

intuitionistic case and lift to first order without being rebuilt. The framework in which the proposal can be carried out for linear logic is order algebraizability over po-varieties, with the polarity fixed by the connectives, since the poset collapse of a symmetric monoidal closed category is a partially ordered monoid and not a lattice.

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